Integrale curbilinii de speța I

- Să se calculeze următoarele integrale curbilinii de prima speță:

1. $\int_\gamma (x^2+y^2)\ln z\, ds$

$$\gamma(t): \begin{cases} x(t) = e^t \cos t \\ y(t) = e^t \sin t \\ z(t) = e^t \end{cases}, \quad t \in [0,1]$$

$f(x,y,z) = (x^2+y^2)\ln z \Rightarrow f(\gamma(t)) = t e^{2t}$

$$\gamma'(t): \begin{cases} x'(t) = e^t(\cos t - \sin t) \\ y'(t) = e^t(\sin t + \cos t) \\ z'(t) = e^t \end{cases}$$

$\|\gamma'(t)\| = \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} = \sqrt{3}\, e^t$

$$\int_\gamma f\, ds = \int_0^1 f(\gamma(t)) \cdot \|\gamma(t)\|\, dt = \int_0^1 \sqrt{3}\, t e^{2t}\, dt = \frac{\sqrt{3}}{2}\int_0^1 (e^{2t})' \cdot t\, dt = \frac{\sqrt{3}}{2} e^{2t}\cdot t\Big|_0^1 - \frac{\sqrt{3}}{4} e^{2t}\Big|_0^1 = \frac{2\sqrt{3}e^2 + \sqrt{3}}{4}$$

2. $\int_\gamma y e^{-x}\, ds$

$$\gamma(t): \begin{cases} x(t) = \ln(1+t^2) \\ y(t) = 2\operatorname{arctg} t - t \end{cases}, \quad t \in [0,1]$$

$f(x,y) = y e^{-x} \Rightarrow f(\gamma(t)) = (2\operatorname{arctg} t - t)\cdot \frac{1}{1+t^2}$

$$\gamma'(t): \begin{cases} x'(t) = \frac{2t}{1+t^2} \\ y'(t) = \frac{2}{1+t^2} - 1 \end{cases}$$

$\|\gamma'(t)\| = \sqrt{(x'(t))^2 + (y'(t))^2} = t$

$$\int_\gamma f\, ds = \int_0^1 f(\gamma(t)) \cdot \|\gamma'(t)\|\, dt = \int_0^1 2\operatorname{arctg} t \cdot (\operatorname{arctg} t)'\, dt - \frac{1}{2}\int_0^1 \frac{2t}{1+t^2}\, dt = \operatorname{arctg}^2 t\Big|_0^1 - \frac{1}{2}\ln(1+t^2)\Big|_0^1 = \frac{\pi^2}{16} - \frac{1}{2}\ln 2$$

3. $\int_\gamma xyz\, ds$

$$\gamma(t): \begin{cases} x(t) = t \\ y(t) = \frac{2}{3}\sqrt{2t^3} \\ z(t) = \frac{1}{2}t^2 \end{cases}, \quad t \in [0,1]$$

$f(x,y,z) = xyz \Rightarrow f(\gamma(t)) = \frac{1}{3}t^3\sqrt{2t^3}$

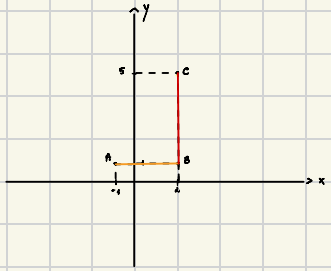
$$\gamma'(t): \begin{cases} x'(t) = 1 \\ y'(t) = \frac{2\cdot 3t^2}{2\sqrt{2t^3}} = \sqrt{2t} \\ z'(t) = t \end{cases}$$

$\|\gamma'(t)\| = \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2} = t+1$

$$\int_\gamma f\, ds = \int_0^1 f(\gamma(t)) \cdot \|\gamma'(t)\|\, dt = \frac{\sqrt{2}}{3}\int_0^1 \sqrt{t^3}(t+1)\, dt = \frac{\sqrt{2}}{3}\int_0^1 (t^{\frac{3}{2}} + t^{\frac{5}{2}})\, dt = \frac{16\sqrt{2}}{105}$$

4. $\int_\gamma x^2 y \, ds$

$A(-1,1)$, $B(2,1)$, $C(2,5)$



[AB] $\gamma_1(t): \begin{cases} x(t) = 3t - 1 \\ y(t) = 1 \end{cases}$, $t \in [0,1]$, ou $\|\gamma_1'(t)\| = 3$

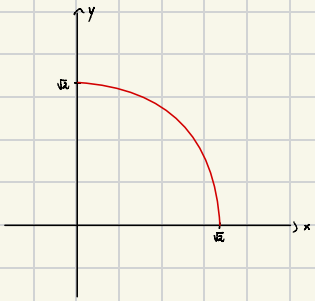
[BC] $\gamma_2(t): \begin{cases} x(t) = 2 \\ y(t) = 1 + 4t \end{cases}$, $t \in [0,1]$, ou $\|\gamma_2'(t)\| = 4$

$f(x,y) = x^2 y \Rightarrow f(\gamma_1(t)) = (3t-1)^2$

$f(\gamma_2(t)) = 4(1+4t)$

$$\int_\gamma f \, ds = \int_{\gamma_1} f \, ds + \int_{\gamma_2} f \, ds = 3\int_0^1 (9t^2 - 6t + 1)\, dt + 16\int_0^1 (1+4t)\, dt = 51$$

5. $\int_\gamma x^2 ds$, $\gamma: x^2 + y^2 = 2$, $x \geq 0$, $y \geq 0$



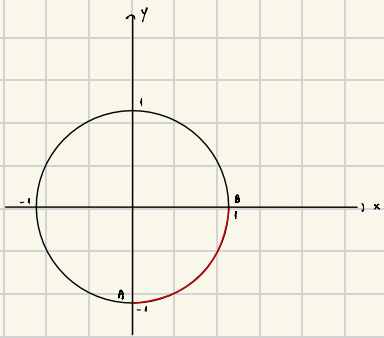
$\gamma(t): \begin{cases} x(t) = \sqrt{2}\cos t \\ y(t) = \sqrt{2}\sin t \end{cases}$, $t \in \left[0, \frac{\pi}{2}\right]$

$f(x,y) = x^2 \Rightarrow f(\gamma(t)) = 2\cos^2 t$

$\|\gamma'(t)\| = \sqrt{(x'(t))^2 + (y'(t))^2} = \sqrt{2}$

$$\int_\gamma f \, ds = \int_0^{\pi/2} 2\sqrt{2}\cos^2 t \, dt = 2\sqrt{2}\int_0^{\pi/2} \frac{1+\cos 2t}{2}\, dt = \frac{\sqrt{2}\,\pi}{2}$$

6. $\int_\gamma (x^2 + y^2)\, ds$, γ: arco de cerc $x^2 + y^2 = 1$, de la $A(0,-1)$ la $B(1,0)$



$\gamma(t): \begin{cases} x(t) = \cos t \\ y(t) = \sin t \end{cases}$, $t \in \left[-\frac{\pi}{2}, 0\right]$

$\|\gamma'(t)\| = 1$

$f(x,y) = x^2 + y^2 \Rightarrow f(\gamma(t)) = 1$

$$\int_\gamma f \, ds = \int_{-\pi/2}^{0} f(\gamma(t)) \cdot \|\gamma'(t)\| \, dt = \frac{\pi}{2}$$

7. $\int_\gamma (x + y)\, ds$

$\gamma(t): \begin{cases} x(t) = e^t + 1 \\ y(t) = e^t - 1 \end{cases}$, $t \in [0, \ln 2]$

$f(x,y) = x + y \Rightarrow f(\gamma(t)) = 2e^t$

$\|\gamma'(t)\| = \sqrt{2}\, e^t$

$$\int_\gamma f \, ds = \int_0^{\ln 2} f(\gamma(t)) \cdot \|\gamma'(t)\| \, dt = \sqrt{2}\int_0^{\ln 2} 2e^{2t}\, dt = \sqrt{2}\, e^{2t}\Big|_0^{\ln 2} = 3\sqrt{2}$$

8. $\int_\gamma (2x-y)\,ds$

$\gamma(t)$: $\begin{cases} x(t) = \sin t \\ y(t) = \cos t \end{cases}$, $t \in \left[0, \frac{\pi}{2}\right]$

$f(x,y) = 2x - y \Rightarrow f(\gamma(t)) = 2\sin t - \cos t$

$\|\gamma'(t)\| = 1$

$\int_\gamma f\,ds = \int_0^{\pi/2} (2\sin t - \cos t)\,dt = 1$

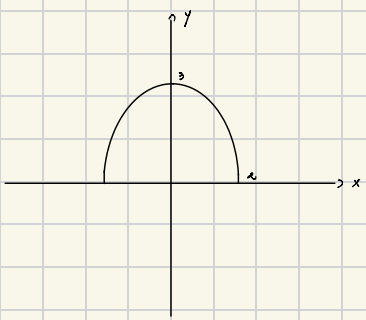
9. $\int_\gamma xyz\,ds$, γ: segmentul AB, cu $A(1,-1,2)$ și $B(3,2,5)$

$\gamma(t)$: $\begin{cases} x(t) = 1+2t \\ y(t) = 3t-1 \\ z(t) = 2+3t \end{cases}$, $t \in [0,1]$, $\|\gamma'(t)\| = \sqrt{22}$

$f(x,y,z) = xyz \Rightarrow f(\gamma(t)) = (1+2t)(3t-1)(2+3t) = (6t^2+t-1)(2+3t) = 18t^3+15t^2-t-2$

$\int_\gamma f\,ds = \int_0^1 f(\gamma(t)) \cdot \|\gamma'(t)\|\,dt = 7\sqrt{22}$

10. $\int_\gamma xy\,ds$, γ: $\frac{x^2}{4} + \frac{y^2}{9} = 1$, $y \geq 0$



$\gamma(t)$: $\begin{cases} x(t) = 2\cos t \\ y(t) = 3\sin t \end{cases}$, $t \in [0, \pi]$

$\|\gamma'(t)\| = \sqrt{4+5\cos^2 t}$

$f(x,y) = xy \Rightarrow f(\gamma(t)) = 6\sin t\cos t$

$\int_\gamma f\,ds = \int_0^\pi 6\sin t\cos t\sqrt{4+5\cos^2 t}\,dt = -\frac{3}{5}\int_9^9 \sqrt{u}\,du = 0$

$4+5\cos^2 t = u \Rightarrow -10\sin t\cos t\,dt = du \Rightarrow 6\sin t\cos t\,dt = -\frac{3}{5}\,du$

$t = 0 \Rightarrow u = 9$

$t = \pi \Rightarrow u = 9$

11. $\int_\gamma (x^2+y^2)\,ds$

$\gamma(t)$: $\begin{cases} x(t) = 4t-1 \\ y(t) = 3t+1 \end{cases}$, $t \in [-1,1]$

$f(x,y) = x^2+y^2 \Rightarrow f(\gamma(t)) = 25t^2 - 2t + 2$

$\|\gamma'(t)\| = 5$

$\int_\gamma f\,ds = \int_{-1}^1 f(\gamma(t)) \cdot \|\gamma'(t)\|\,dt = \frac{66}{3} \cdot 5 = \frac{210}{3}$

12. $\int_\gamma y^2\,ds$, $\gamma: x^2+y^2=4$, $x\le 0$, $y\ge 0$

$\gamma(t): \begin{cases} x(t)=2\cos t \\ y(t)=2\sin t \end{cases}$, $t\in\left[\frac{\pi}{2},\pi\right]$

$\|\gamma'(t)\| = 2$

$f(x,y)=y^2 \Rightarrow f(\gamma(t)) = 4\sin^2 t$

$$\int_\gamma f\,ds = \int_{\pi/2}^{\pi} f(\gamma(t))\cdot\|\gamma'(t)\|\,dt = 8\int_{\pi/2}^{\pi}\sin^2 t\,dt = 8\int_{\pi/2}^{\pi}\frac{1-\cos 2t}{2}\,dt = 4\int_{\pi/2}^{\pi}dt - 2\int_{\pi/2}^{\pi}2\cos 2t\,dt = 2\pi - 0 = 2\pi$$

- Să se calculeze lungimea următoarelor drumuri:

13. $\gamma(t): \begin{cases} x(t)=e^t \\ y(t)=e^{\frac{3}{2}t} \end{cases}$, $t\in[0,1]$

$\gamma'(t): \begin{cases} x'(t)=e^t \\ y'(t)=\frac{3}{2}e^{\frac{3}{2}t} \end{cases}$

$\|\gamma'(t)\| = \sqrt{(x'(t))^2+(y'(t))^2} = \frac{1}{2}e^t\sqrt{4+9e^t}$

$$L = \int_\gamma ds = \int_0^1 \|\gamma'(t)\|\,dt = \frac{1}{18}\int_0^1 9e^t\sqrt{4+9e^t}\,dt = \frac{1}{18}\int_{13}^{4+9e} u^{\frac{1}{2}}\,du = \frac{1}{27}u^{\frac{3}{2}}\Big|_{13}^{4+9e} = \frac{1}{27}\left((4+9e)^{\frac{3}{2}} - 13^{\frac{3}{2}}\right)$$

$4+9e^t = u \Rightarrow 9e^t\,dt = du$

$t=0 \Rightarrow u=13$

$t=1 \Rightarrow u=4+9e$

14. $\gamma: y=\ln(\sin x)$, $x\in\left[\frac{\pi}{6},\frac{\pi}{3}\right]$

$\gamma(t): \begin{cases} x(t)=t \\ y(t)=\ln(\sin t) \end{cases}$, $t\in\left[\frac{\pi}{6},\frac{\pi}{3}\right]$

$\gamma'(t): \begin{cases} x'(t)=1 \\ y'(t)=\frac{\cos t}{\sin t} \end{cases}$

$\|\gamma'(t)\| = \frac{1}{\sin t}$

$$L = \int_\gamma ds = \int_{\pi/6}^{\pi/3}\|\gamma'(t)\|\,dt = \int_{\pi/6}^{\pi/3}\frac{1}{\sin t}\,dt = \int_{\pi/6}^{\pi/3}\frac{\sin t}{\sin^2 t}\,dt = \int_{\pi/6}^{\pi/3}\frac{\sin t}{1-\cos^2 t}\,dt$$

$\cos t = u \Rightarrow -\sin t\,dt = du$

$t=\frac{\pi}{6} \Rightarrow u=\frac{\sqrt{3}}{2}$

$t=\frac{\pi}{3} \Rightarrow u=\frac{1}{2}$

$$L = \int_{1/2}^{\sqrt{3}/2}\frac{1}{1-u^2}\,du = -\int_{1/2}^{\sqrt{3}/2}\frac{1}{u^2-1}\,du = -\frac{1}{2}\ln\left|\frac{u-1}{u+1}\right|\Big|_{1/2}^{\sqrt{3}/2} = \frac{1}{2}\left(\ln 3 + \ln\left(\frac{2-\sqrt{3}}{2+\sqrt{3}}\right)\right)$$

15. γ: segmentul AB, cu $A(1,2)$ și $B(3,5)$

$$r(t): \begin{cases} x(t) = 1+2t \\ y(t) = 2+3t \end{cases}, \quad t \in [0,1]$$

$$\|r'(t)\| = \sqrt{13}$$

$$L = \int_\gamma ds = \int_0^1 \sqrt{13}\, dt = \sqrt{13}$$

16. γ: $r = e^{at}$, $a \in \mathbb{R}^*$, $t \in [0, 2\pi]$ (arcul spiralei logaritmice)

$$r(t): \begin{cases} x(t) = r\cos t \\ y(t) = r\sin t \end{cases}, \ t \in [0,2\pi] \iff r(t): \begin{cases} x(t) = e^{at}\cos t \\ y(t) = e^{at}\sin t \end{cases}, \ t \in [0,2\pi]$$

$$r'(t): \begin{cases} x'(t) = e^{at}(a\cos t - \sin t) \\ y'(t) = e^{at}(a\sin t + \cos t) \end{cases}$$

$$\|r'(t)\| = e^{at}\sqrt{a^2+1}$$

$$L = \int_\gamma ds = \int_0^{2\pi} \|r'(t)\|\, dt = \sqrt{a^2+1}\cdot\frac{1}{a}\int_0^{2\pi} (at)'\, e^{at}\, dt = \frac{e^{2\pi a}-1}{a}\sqrt{a^2+1}$$

17 $\rho(x,y,z) = 1 + x + z$

$$r(t): \begin{cases} x(t) = \cos t \\ y(t) = \sin t \\ z(t) = t \end{cases}, \ t \in [0,2\pi] \quad \text{(arc de elice)}$$

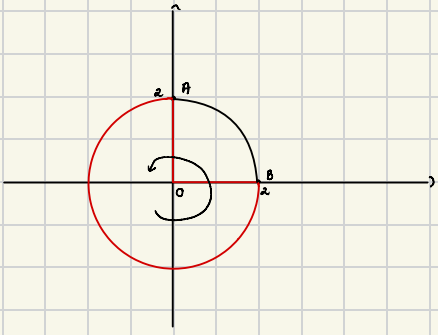
$$\|r'(t)\| = \sqrt{2}$$

$$\rho(r(t)) = 1 + t + \cos t$$

$$m = \int_\gamma \rho(x,y,z)\, ds = \int_0^{2\pi} \rho(r(t))\cdot\|r'(t)\|\, dt = \sqrt{2}\int_0^{2\pi} (1 + t + \cos 2t)\, dt = 2\sqrt{2}\pi(1+\pi)$$

18 $\rho = 1$

γ: arcul de cerc $x^2 + y^2 = 4$, între punctele $A(0,2)$ și $B(2,0)$ și segmentele AO și OB, cu $O(0,0)$ (sens trigonometric)



$$\gamma_1(t): \begin{cases} x(t) = 2\cos t \\ y(t) = 2\sin t \end{cases}, \ t \in \left[\frac{3\pi}{2}, 2\pi\right] \quad (\text{arcul de cerc din } \textit{IV}) \quad \Rightarrow \|\gamma_1'(t)\| = 2$$

$$\gamma_2(t): \begin{cases} x(t) = 0 \\ y(t) = 2 - 2t \end{cases}, \ t \in [0,1] \quad (\text{segmentul } [AO]) \quad \Rightarrow \|\gamma_2'(t)\| = 2$$

$$\gamma_3(t): \begin{cases} x(t) = 2t \\ y(t) = 0 \end{cases}, \ t \in [0,1] \quad (\text{segmentul } [OB]) \quad \Rightarrow \|\gamma_3'(t)\| = 2$$

$$m = \int_\gamma \rho \, ds = \int_\gamma ds = \int_{\gamma_1} ds + \int_{\gamma_2} ds + \int_{\gamma_3} ds = 3\pi + 4$$

$$\int_{\gamma_1} ds = \int_{\frac{3\pi}{2}}^{2\pi} \|\gamma_1'(t)\| \, dt = 3\pi$$

$$\int_{\gamma_2} ds = \int_0^1 \|\gamma_2'(t)\| \, dt = 2$$

$$\int_{\gamma_3} ds = \int_0^1 \|\gamma_3'(t)\| \, dt = 2$$

$$x_G = \frac{1}{m}\int_\gamma x\,\rho(x,y,z)\,ds = \frac{1}{m}\left(\int_{\gamma_1} x\,ds + \int_{\gamma_2} x\,ds + \int_{\gamma_3} x\,ds\right) = \frac{1}{3\pi+4}\left(\int_{\frac{3\pi}{2}}^{2\pi} 4\cos t\,dt + \int_0^1 0\,dt + \int_0^1 4t\,dt\right) = -\frac{2}{4+3\pi}$$

$$y_G = \frac{1}{m}\int_\gamma y\,\rho(x,y,z)\,ds = \frac{1}{m}\left(\int_{\gamma_1} y\,ds + \int_{\gamma_2} y\,ds + \int_{\gamma_3} y\,ds\right) = \frac{1}{3\pi+4}\left(\int_{\frac{3\pi}{2}}^{2\pi} 4\sin t\,dt + \int_0^1 4(1-t)\,dt + \int_0^1 0\,dt\right) = -\frac{2}{4+3\pi}$$